

Iterative Model Free Safe Exploration using Event-triggered Impulsive Perturbations - Appendix

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1. Appendix A

A.1 Proof of Lemma 1

- 1) This condition is proved by its own definition.
- 2) This condition is proved by having the value function constructed as a sum of positive definite functions $r(\cdot, \cdot)$.
- 3) Let $J_{C_0}(\mathbf{x}_{k+1}) - J_{C_0}(\mathbf{x}_k) < 0 \Leftrightarrow J_{C_0}(\mathbf{x}_{k+1}) < J_{C_0}(\mathbf{x}_k) \Leftrightarrow J_{C_0}(\mathbf{x}_{k+1}) < r(\mathbf{x}_k, C_0(\mathbf{x}_k)) + \gamma J_{C_0}(\mathbf{x}_{k+1}) \Leftrightarrow J_{C_0}(\mathbf{x}_{k+1}) - \gamma J_{C_0}(\mathbf{x}_{k+1}) < r(\mathbf{x}_k, C_0(\mathbf{x}_k)) \Leftrightarrow (1 - \gamma) < \frac{r(\mathbf{x}_k, C_0(\mathbf{x}_k))}{J_{C_0}(\mathbf{x}_{k+1})} \Leftrightarrow \gamma > \frac{J_{C_0}(\mathbf{x}_{k+1}) - r(\mathbf{x}_k, C_0(\mathbf{x}_k))}{J_{C_0}(\mathbf{x}_{k+1})}$. (18)

This proves the lemma.

A.2 Proof of Lemma 2

We first establish the lower bound $\alpha(\|\mathbf{x}_k - \mathbf{x}_k^G\|) \leq J_{C_0}(\mathbf{x}_k)$. With $r(\cdot, \cdot)$ a positive definite function, there exist $\lambda_{min} > 0$ such that $\lambda_{min}\|\mathbf{x}_k - \mathbf{x}_k^G\| \leq r(\mathbf{x}_k, C_0(\mathbf{x}_k))$. Having $r(\mathbf{x}_k, C_0(\mathbf{x}_k)) \leq J_{C_0}(\mathbf{x}_k)$ from (8), we get $\lambda_{min}\|\mathbf{x}_k - \mathbf{x}_k^G\| \leq J_{C_0}(\mathbf{x}_k)$, which defines our continuous $\mathcal{K}$ -class function $\lambda_{min}\|\mathbf{x}_k - \mathbf{x}_k^G\| := \alpha(\|\mathbf{x}_k - \mathbf{x}_k^G\|)$.

To prove the upper bound $J_{C_0}(\mathbf{x}_k) \leq \beta(\|\mathbf{x}_k - \mathbf{x}_k^G\|)$, we use condition 3) from Lemma 1 together with (8), to get

$$\begin{aligned} J_{C_0}(\mathbf{x}_k) &= r(\mathbf{x}_k, C_0(\mathbf{x}_k)) + \gamma J_{C_0}(\mathbf{x}_{k+1}) \\ &\leq r(\mathbf{x}_k, C_0(\mathbf{x}_k)) + \gamma J_{C_0}(\mathbf{x}_k) \\ &\Leftrightarrow (1 - \gamma)J_{C_0}(\mathbf{x}_k) \leq r(\mathbf{x}_k, C_0(\mathbf{x}_k)) \\ &\Leftrightarrow J_{C_0}(\mathbf{x}_k) \\ &\leq \frac{1}{(1 - \gamma)} r(\mathbf{x}_k, C_0(\mathbf{x}_k)). \end{aligned} \quad (19)$$

There exists $\lambda_{max} > 0$ such that $r(\mathbf{x}_k, C_0(\mathbf{x}_k)) \leq \lambda_{max}\|\mathbf{x}_k - \mathbf{x}_k^G\|$. Having $J_{C_0}(\mathbf{x}_k) \leq (1/(1 - \gamma))r(\mathbf{x}_k, C_0(\mathbf{x}_k))$ from (19), we get $J_{C_0}(\mathbf{x}_k) \leq (\lambda_{max}/(1 - \gamma))\|\mathbf{x}_k - \mathbf{x}_k^G\|$, which defines our continuous $\mathcal{K}$ -class function $(\lambda_{max}/(1 - \gamma))\|\mathbf{x}_k - \mathbf{x}_k^G\| := \beta(\|\mathbf{x}_k - \mathbf{x}_k^G\|)$. This proves the lemma.

A.3 Proof of Theorem 1

Let $k_j^+ = k_j + 1$ be the first time sample after a perturbation impulse that makes $J_{C_0}(\mathbf{x}_k) > a(1 - b)^k$, and $k_j^- = k_j$ be the

first time sample after k_j^+ such that $J_{C_0}(\mathbf{x}_k) < a(1 - b)^k$. A potential trajectory of system (1) is depicted in Fig. 7 for demonstration purposes. The first impulse time k_1 is shown to happen at time $k_1 = 2$, where $J_{C_0}(\mathbf{x}_{k_1}) < a(1 - b)^{k_1}$. The impulse time k_1 destabilizes the system, having thus time step k_1^+ such that $J_{C_0}(\mathbf{x}_{k_1^+}) > a(1 - b)^{k_1^+}$. Namely,

$$k_1^+ = \min\{k \in (k_1, \infty): J_{C_0}(\mathbf{x}_k) > a(1 - b)^k\}. \quad (20)$$

After k_1^+ , the system evolves p -steps based on $F(\mathbf{x}_k, C_0(\mathbf{x}_k))$, until $J_{C_0}(\mathbf{x}_k) < a(1 - b)^k$ for $k = k_1^-$. Let

$$k_1^- = \min\{k \in (k_1^+, \infty): J_{C_0}(\mathbf{x}_k) < a(1 - b)^k\}. \quad (21)$$

Then, based on Assumption 2 we have

$$\begin{aligned} J_{C_0}(\mathbf{x}_{k_1^-}) &\leq c^p J_{C_0}(\mathbf{x}_{k_1^+}) \leq mc^p J_{C_0}(\mathbf{x}_{k_1}) \leq mc^p a(1 - b)^{k_1} \\ &= mc^p a(1 - b)^{k_1} \frac{(1 - b)^{k_1^- - k_1}}{(1 - b)^{k_1^- - k_1}} \\ &= \frac{mc^p}{(1 - b)^{k_1^- - k_1}} a(1 - b)^{k_1^-}. \end{aligned} \quad (22)$$

Therefore, if $mc^p/(1 - b)^{k_1^- - k_1} < 1$, then the value of $J_{C_0}(\mathbf{x}_{k_1^-})$ does not exceed $a(1 - b)^{k_1^-}$ at time k_1^- . This is equivalent of saying that the effect of the perturbation impulse from time k_1 is damped at time k_1^- under rule (11). For any general noise event time k_j , we have $k_j^+ = \min\{k \in (k_j, \infty): J_{C_0}(\mathbf{x}_k) > a(1 - b)^k\}$ and $k_j^- = \min\{k \in (k_j^+, \infty): J_{C_0}(\mathbf{x}_k) < a(1 - b)^k\}$. Using the same proof steps as above, we can conclude that $J_{C_0}(\mathbf{x}_{k_j^-}) \leq \frac{mc^p}{(1 - b)^{k_j^- - k_j}} a(1 - b)^{k_j^-}$, showing that $J_{C_0}(\mathbf{x}_{k_j^-})$ settles below the threshold $a(1 - b)^{k_j^-}$ after the exploratory noise at time k_j . As $a(1 - b)^k \rightarrow 0$ we have $J_{C_0}(\mathbf{x}_k) \leq a(1 - b)^k \rightarrow 0$, proving that the system in closed loop with control rule (11) is both stable and attractive. Then, based on Lemma 2, we get $\|\mathbf{x}_k - \mathbf{x}_k^G\| \rightarrow 0 \Leftrightarrow \mathbf{x}_k \rightarrow \mathbf{x}_k^G$, showing that the system is asymptotically stable to $\mathbf{x}_k^G$ in Ω_X under (11), as $k \rightarrow \infty$.

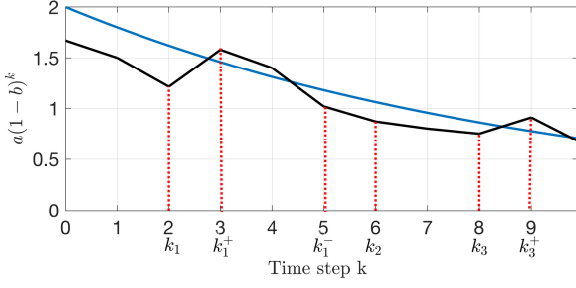


Figure 7. Visual presentation of J_{C_0} function evolution scenario for $T_s = 1$ and $\varrho = 2$: target Lyapunov signal (blue); $J_{C_0}(\mathbf{x}_k)$ (dark). The impulse times generated by (11) are $\{k_1, k_2, k_3\}$: k_1 and k_3 are destabilizing the system, while the one from k_2 is stabilizable.

A.4 Proof of Lemma 3

Let $\mathbf{x}_{k_j+1}^l = F(\mathbf{x}_{k_j}, C_0(\mathbf{x}_{k_j}) + \mu_l(\mathbf{x}_{k_j}, J_{C_0}(\mathbf{x}_{k_j}), V_l))$, and $\mathbf{x}_{k_j} \in \Psi$. The Lipschitz inequality during the additive impulsive perturbation is defined as

$$\begin{aligned} \|\mathbf{x}_{k_j+1}^{l+1} - \mathbf{x}_{k_j+1}^l\| &\leq L_F \|C_0(\mathbf{x}_{k_j}) + \mu_{l+1}(\mathbf{x}_{k_j}, J_{C_0}(\mathbf{x}_{k_j}), V_l) \\ &\quad - C_0(\mathbf{x}_{k_j}) + \mu_l(\mathbf{x}_{k_j}, J_{C_0}(\mathbf{x}_{k_j}), V_l)\| = \\ L_F \|\mu_{l+1}(\mathbf{x}_{k_j}, J_{C_0}(\mathbf{x}_{k_j}), V_l) - \mu_l(\mathbf{x}_{k_j}, J_{C_0}(\mathbf{x}_{k_j}), V_l)\|. \end{aligned} \quad (23)$$

For sequential trials l and $l+1$, for a linear variance increase according to $\theta(\cdot)$ we get

$$\|\mu_{l+1}(\mathbf{x}_{k_j}, J_{C_0}(\mathbf{x}_{k_j}), V_l) - \mu_l(\mathbf{x}_{k_j}, J_{C_0}(\mathbf{x}_{k_j}), V_l)\| \leq \varphi. \quad (24)$$

This results in

$$\|\mathbf{x}_{k_j+1}^{l+1} - \mathbf{x}_{k_j+1}^l\| \leq L_F \varphi. \quad (25)$$

Given that $\varphi \leq d_{\min}(\mathbf{x}_{k_j+1}^l, \Xi)/L_F$, we obtain $\|\mathbf{x}_{k_j+1}^{l+1} - \mathbf{x}_{k_j+1}^l\| \leq d_{\min}(\mathbf{x}_{k_j+1}^l, \Xi)$. This proves the lemma.

A.5 Proof of Theorem 2

For any exploration trial l , we define the one-step reachable set for a state $\mathbf{x}_{k_j}$ as the set of all possible transitions from $\mathbf{x}_{k_j}$ under $C_0(\mathbf{x}_{k_j}) + \mu_l(\mathbf{x}_{k_j}, J_{C_0}(\mathbf{x}_{k_j}), V_l)$, namely

$$\begin{aligned} X_l^+(\mathbf{x}_{k_j}) = \\ \left\{ \mathbf{x}_{k_j+1} \mid \mathbf{x}_{k_j+1} = F \left(\mathbf{x}_{k_j}, C_0(\mathbf{x}_{k_j}) + \mu_l(\mathbf{x}_{k_j}, J_{C_0}(\mathbf{x}_{k_j}), V_l) \right) \right\}. \end{aligned} \quad (26)$$

Then, based on the set inclusion Definition 3 ($\Psi \subset B \subset \Xi$ or $\Xi \subset B \subset \Psi$) and on Lemma 1, if $X_l^+(\mathbf{x}_{k_j}) \cap B \neq \emptyset$ we have $X_l^+(\mathbf{x}_{k_j}) \cap \Xi = \emptyset$. Then, according to Algorithm 2, $T_1 = T_{l-1} \cup \{\mathbf{x}_{k_j}\}$, such that for all trials $l+1, \dots, L_{\max}$ we get

$$\begin{aligned} X_{[l+1:L_{\max}]}^+(\mathbf{x}_{k_j}) &= \{\mathbf{x}_{k_j+1} \mid \mathbf{x}_{k_j+1} \\ &= F(\mathbf{x}_{k_j}, C_0(\mathbf{x}_{k_j}) + M_{[l+1:L_{\max}]}(\mathbf{x}_{k_j}))\} \\ &= X_l^+(\mathbf{x}_{k_j}). \end{aligned} \quad (27)$$

This proves that the reachable set for state $\mathbf{x}_{k_j}$ is maintained for all subsequent iterations $l+1, \dots, L_{\max}$, avoiding transitions to the unsafe set Ξ . Based on the φ -bounded increase in exploration controller variance between successive l trials, this property extends to all $\mathbf{x}_k \in \Psi$ as $l \rightarrow L_{\max}$, proving that system (14) remains invariant in $\Psi \cup B$.

2. Appendix B

B.1 Approximate Q-learning algorithm

Arriving at the optimal cost function is achieved in an iterative manner, using the Q-learning algorithm as next described.

Algorithm 1. Q-learning

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- Step 1. Initialize $Q_0(\mathbf{x}_k, \mathbf{u}_k)$, $C_0(\mathbf{x}_{k+1})$ and $i = 0$.
- Step 2. Update the Q-function using $Q_{i+1}(\mathbf{x}_k, \mathbf{u}_k) = r(\mathbf{x}_k, \mathbf{u}_k) + \gamma Q_i(\mathbf{x}_{k+1}, C_i(\mathbf{x}_{k+1}))$.
- Step 3. Update the controller using $\mathbf{u}_k = C_{i+1}(\mathbf{x}_k) = \text{argmin}_{\mathbf{C}} \{Q_{i+1}(\mathbf{x}_k, \mathbf{u}_k)\}$.
- Step 4. If no changes happen between $Q_{i+1}(\mathbf{x}_k, \mathbf{u}_k)$ and $Q_i(\mathbf{x}_k, \mathbf{u}_k)$, stop the run and get $C_i(\mathbf{x}_k)$. Else, go to Step 2.
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B.2 Approximate Q-learning algorithm

Using the database DBM_l collected during Algorithm 2, we update the Q-function using

$$\mathbf{w}_Q^{i+1} = \arg \min_{\mathbf{w}} \frac{1}{\Pi} \sum_{k=0}^{\Pi-1} \left(\tilde{Q}(\mathbf{x}_k, \mathbf{u}_k, \mathbf{w}) - r(\mathbf{x}_k, \mathbf{u}_k) - \gamma \tilde{Q}(\mathbf{x}_{k+1}, \tilde{C}(\mathbf{x}_{k+1}, \mathbf{w}_C^i), \mathbf{w}_Q^i) \right)^2 \quad (28)$$

The controller NN weights at iteration i are updated using

$$\mathbf{w}_C^{i+1} = \arg \min_{\mathbf{w}} \frac{1}{\Pi} \sum_{k=0}^{\Pi-1} \tilde{Q}(\mathbf{x}_k, \tilde{C}(\mathbf{x}_k, \mathbf{w}), \mathbf{w}_Q^{i+1}). \quad (29)$$

The Approximate Q-learning (AQL) algorithm is described by the following steps.

Algorithm 3. Approximate Q-learning

Input: Collected dataset DBM_l for trial l .

- Step 1. Initialize $l_{\mathbf{w}_C}, I, \Delta_Q, \iota = 0, i = 0$.
- Step 2. Update the Q-function using (28).
- Step 3. Initialize $\iota = 0$ and $\mathbf{w}^{0,i} = \mathbf{w}^i$. Issue I gradient steps using (29). Then, set $\mathbf{w}^{i+1} = \mathbf{w}^{l,i}$.
- Step 4. If $\|\mathbf{w}_Q^{i+1} - \mathbf{w}_Q^i\| < \Delta_Q$, stop the algorithm run and use controller $\tilde{C}(\mathbf{x}_k, \mathbf{w}_C^i)$. Else, go to Step 2.
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B.3 ICVF NN approximation

The ICVF NN approximation learning requires a distinct dataset collected using only $C_0(\mathbf{x}_k)$ in closed loop with (1). Let the collected database be $DBM_{C_0} = \{(\mathbf{x}_k, \mathbf{u}_k, \mathbf{x}_{k+1})\}, k = 1:\Pi_{C_0}$, where Π_{C_0} is the number of samples. We issue an iterative process with index $\iota = 1, 2, 3, \dots, I$ and solve

$$\mathbf{w}_{J_{C_0}}^{t+1} = \arg \min_{\mathbf{w}} \frac{1}{\Pi_{C_0}} \sum_{k=0}^{\Pi_{C_0}} \left(\tilde{J}_{C_0, \iota}(\mathbf{x}_k, \mathbf{w}) - r(\mathbf{x}_k, \mathbf{u}_k) - \gamma \tilde{J}_{C_0, \iota}(\mathbf{x}_{k+1}, \mathbf{w}_{J_{C_0}}^t) \right)^2. \quad (30)$$

The Approximate initial controller Value function algorithm is described by the following steps.

Algorithm 4. Approximate initial controller Value Function

Step 1. Initialize $I, \Delta_Q, \Pi_{C_0}, \iota = 0$. Collect database DBM_{C_0} using the controller $C_0(\mathbf{x}_k)$ in closed loop with system (1).

Step 2. Update the Q-function using (30).

Step 3. If $\|\mathbf{w}_Q^{t+1} - \mathbf{w}_Q^t\| < \Delta_{J_{C_0}}$, stop the algorithm run and use $\tilde{J}_{C_0, \iota}(\mathbf{x}_k, \mathbf{w}_{J_{C_0}}^t)$. Else, go to Step 2.

The convergence of this algorithm using NN approximators was devised in Lala (2025).

3. Appendix C

In this section we analyze how the selection of ρ impacts the detection of B and T_l , and consequently the system safety. The minimal radius ensuring reliable detection is characterized by the next theorem.

Theorem 3. Let Λ be a finite set of points and $\rho_{safe} = \max\{\rho > 0 | \forall \mathbf{x} \in \Lambda, \exists \mathbf{y} \in \Lambda: \|\mathbf{x} - \mathbf{y}\| < \rho\}$. Let the open ball $BL(\mathbf{x}, \rho) = \{\mathbf{y} | \|\mathbf{x} - \mathbf{y}\| < \rho\}$. Then, for any $\rho \geq \rho_{safe}$ and query point $\mathbf{x}_k, \mathbf{x}_k \in \Lambda^\rho = \cup_{\mathbf{x} \in \Lambda} BL(\mathbf{x}, \rho)$ if and only if $KNearNS(\mathbf{x}_k, \Lambda, K, \rho_{safe})$ is non-empty.

Proof. By the definition of ρ_{safe} , for any $\mathbf{x} \in \Lambda$ there exists $\mathbf{y} \in \Lambda$ such that $\|\mathbf{x} - \mathbf{y}\| < \rho_{safe} \leq \rho$, with $BL(\mathbf{x}, \rho) \cap BL(\mathbf{y}, \rho) \neq \emptyset$. This creates a chain of overlapping ρ -balls that connects any two points in Λ , making Λ^ρ connected. Now, for any $\mathbf{x}_k \in \Lambda^\rho$, there exists $\mathbf{y} \in T_l$ such that $\|\mathbf{x}_k - \mathbf{y}\| < \rho$, making $KNearNS(\mathbf{x}_k, \Lambda, K, \rho)$ non-empty. Conversely, assume $KNearNS(\mathbf{x}_k, \Lambda, K, \rho)$ is non-empty, then there exists, $\mathbf{y} \in \Lambda$ such that $\|\mathbf{x}_k - \mathbf{y}\| < \rho$, implying that $\mathbf{x}_k \in BL(\mathbf{y}, \rho) \subset \Lambda^\rho$. ■

The practical selection of ρ_{safe} for both finite sets $\Lambda = T_l$ and $\Lambda = B$ is made by starting with a small ρ and increasing it until each point in Λ has a neighbor and Λ^ρ becomes connected through overlapping balls. For increased conservativeness, a

higher value than the found ρ_{safe} can be selected, with the trade-off of limited exploration.

4. Appendix D

D.1 NNs training details

ICVF is approximated using a NN with structure $10 - 100 - 1$. The activation functions are hyperbolic tangent in the hidden layer and linear function at the output layer. Its weights are updated using stochastic gradient run in full-batch mode.

The Q-function and controller are approximated by NNs with architectures of $10 - 50 - 1$ and $8 - 30 - 1$ neurons in each layer. The activation functions are hyperbolic tangent in the hidden layer and linear function at the output layer for both Q-functions and controller NNs. Their weights are updated using stochastic gradient run in full-batch mode.